Machine Learning CS 4641-7641

Neural Networks Forward Pass and Back Propagation

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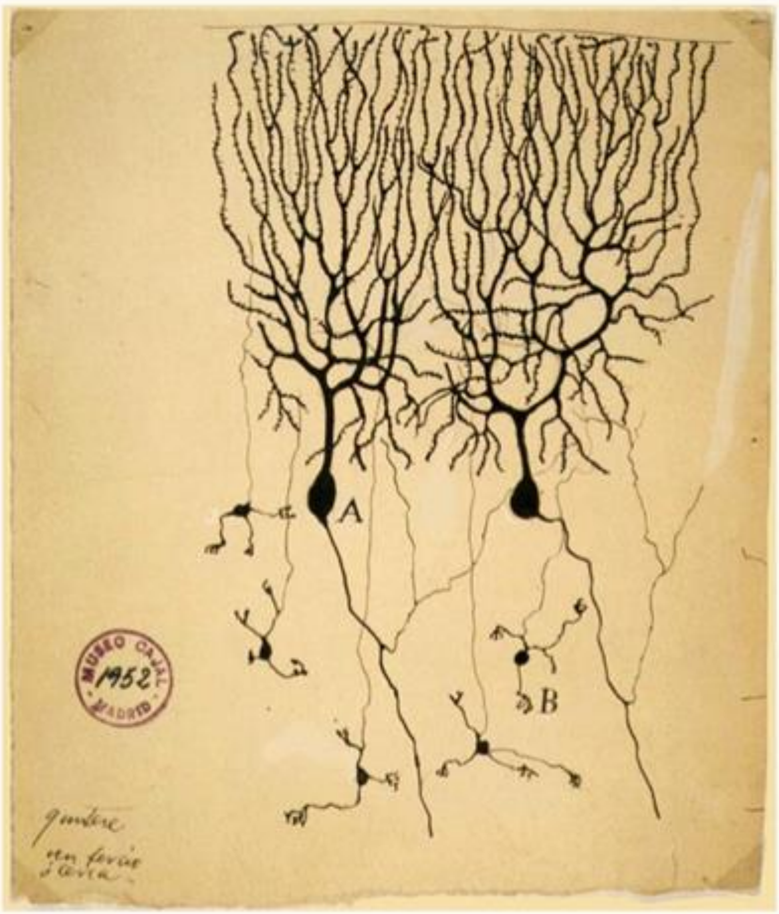
Slides adapted from Mahdi Roozbahani

Inspiration from Biological Neurons



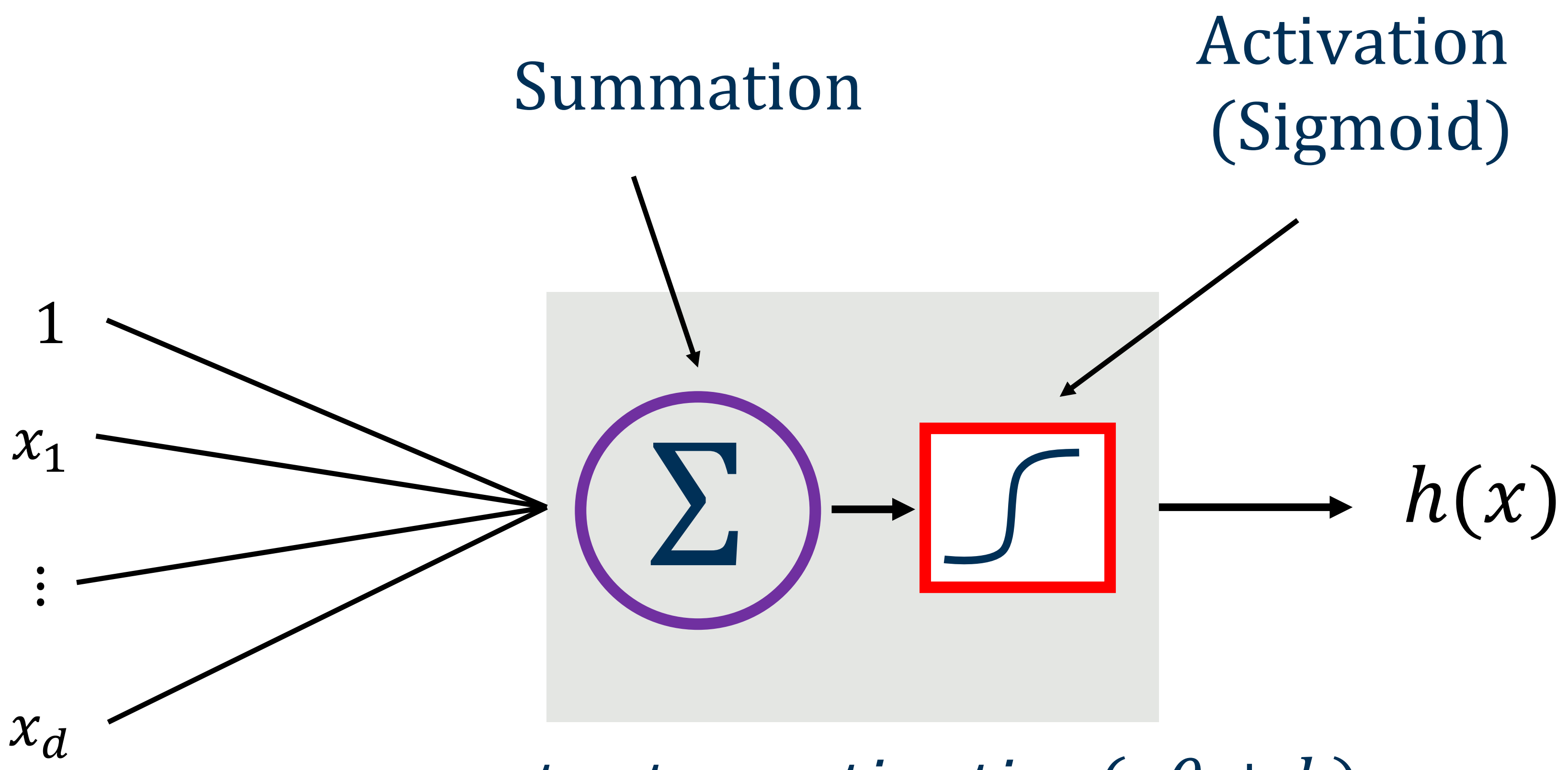
The first drawing of a brain cells by Santiago Ramón y Cajal in 1899

Neurons: core components of brain and the nervous system consisting of

1. Dendrites that collect information from other neurons
2. An axon that generates outgoing spikes

Neural Networks

- A robust approach for approximating **real-valued, discrete-valued or vector valued** functions
- Among the most effective **general purpose** supervised learning methods
- Apt at modeling **complex and hard to interpret data** such as images and audio
- The **backpropagation algorithm** for neural networks has been shown successful in many practical problems:
  - Handwritten character recognition, speech recognition, object recognition, some NLP problems

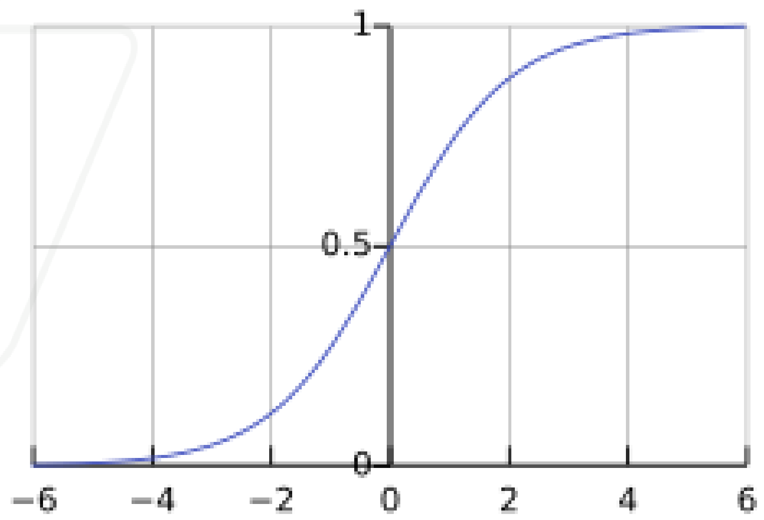


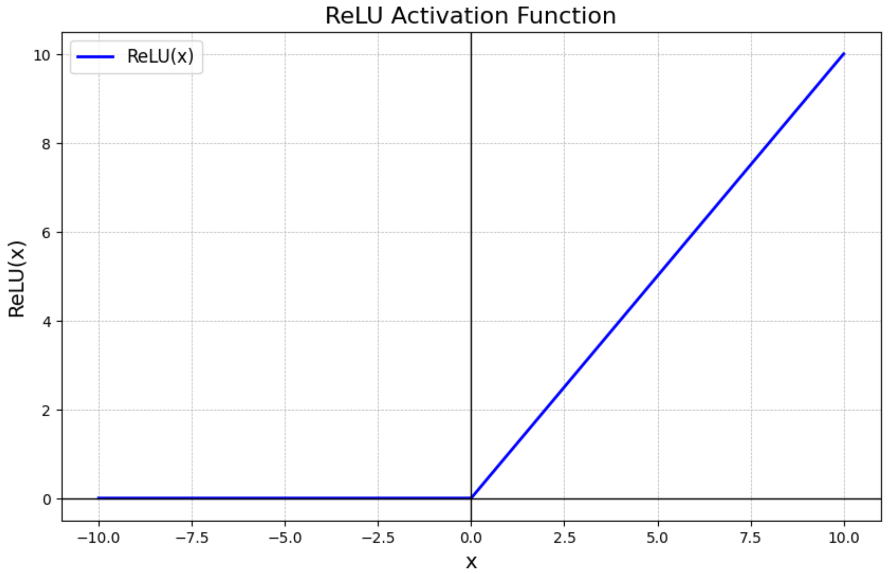
$$output = activation(x\theta + b)$$

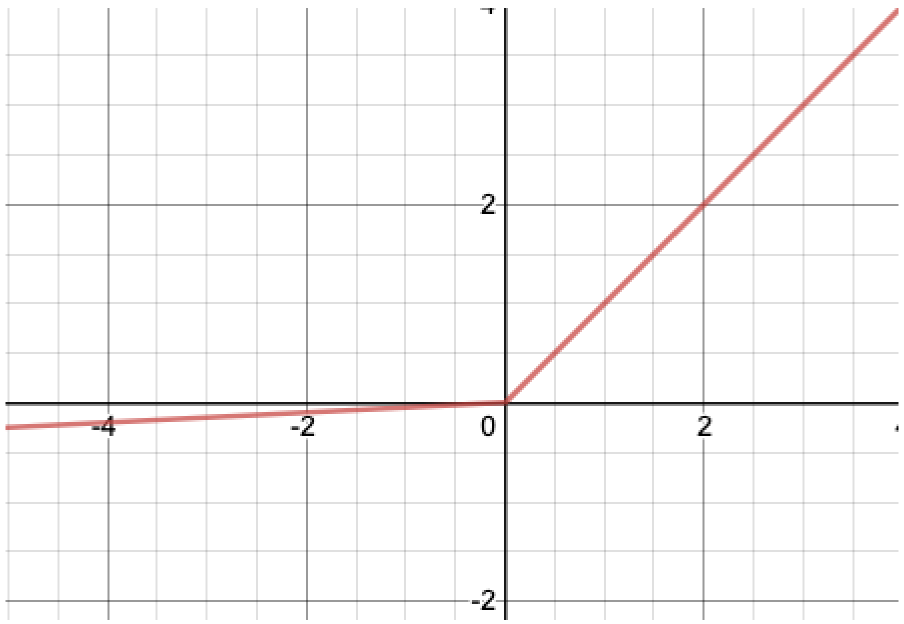
$$h(x) = o(x\theta = u) \rightarrow \text{logistic regression}$$

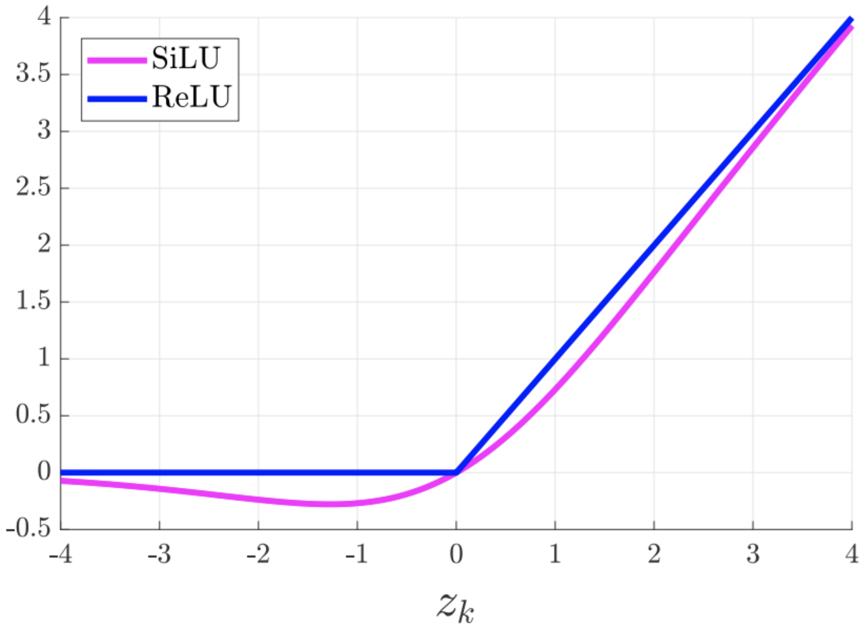
Name of the neuron	Activation function: $activation(z)$
Linear unit	z
Threshold/sign unit	$\text{sgn}(z)$
Sigmoid unit	$\frac{1}{1+\exp(-z)}$
Rectified linear unit (ReLU)	$\max(0, z)$
Tanh unit	$\tanh(z)$

Different activation functions: σ

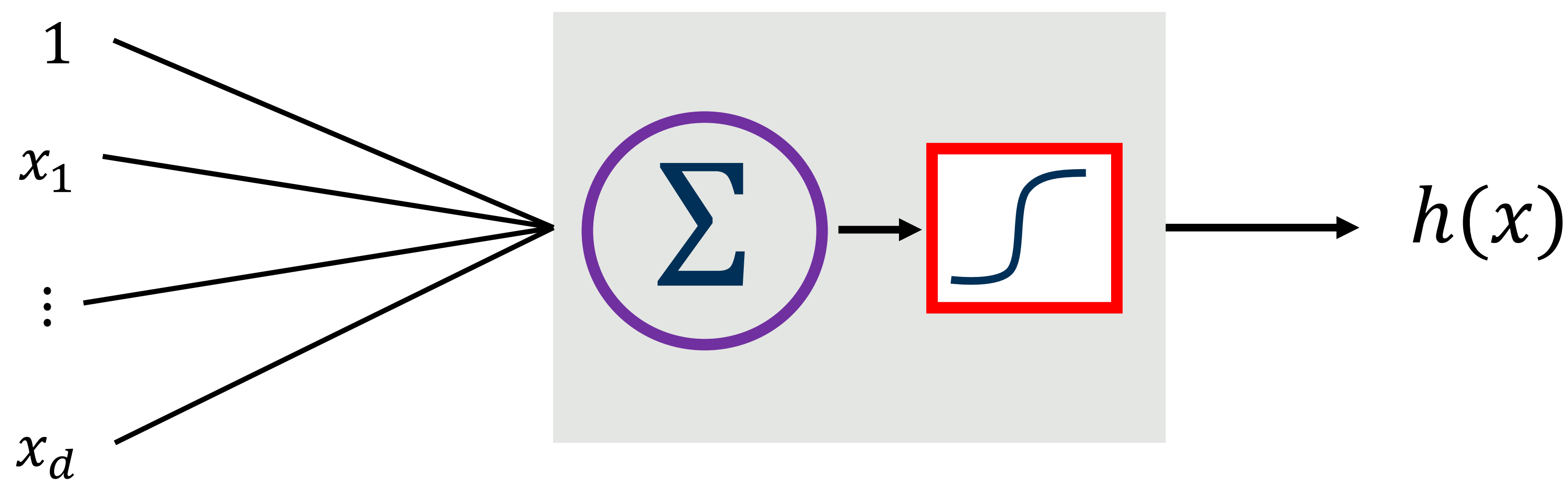








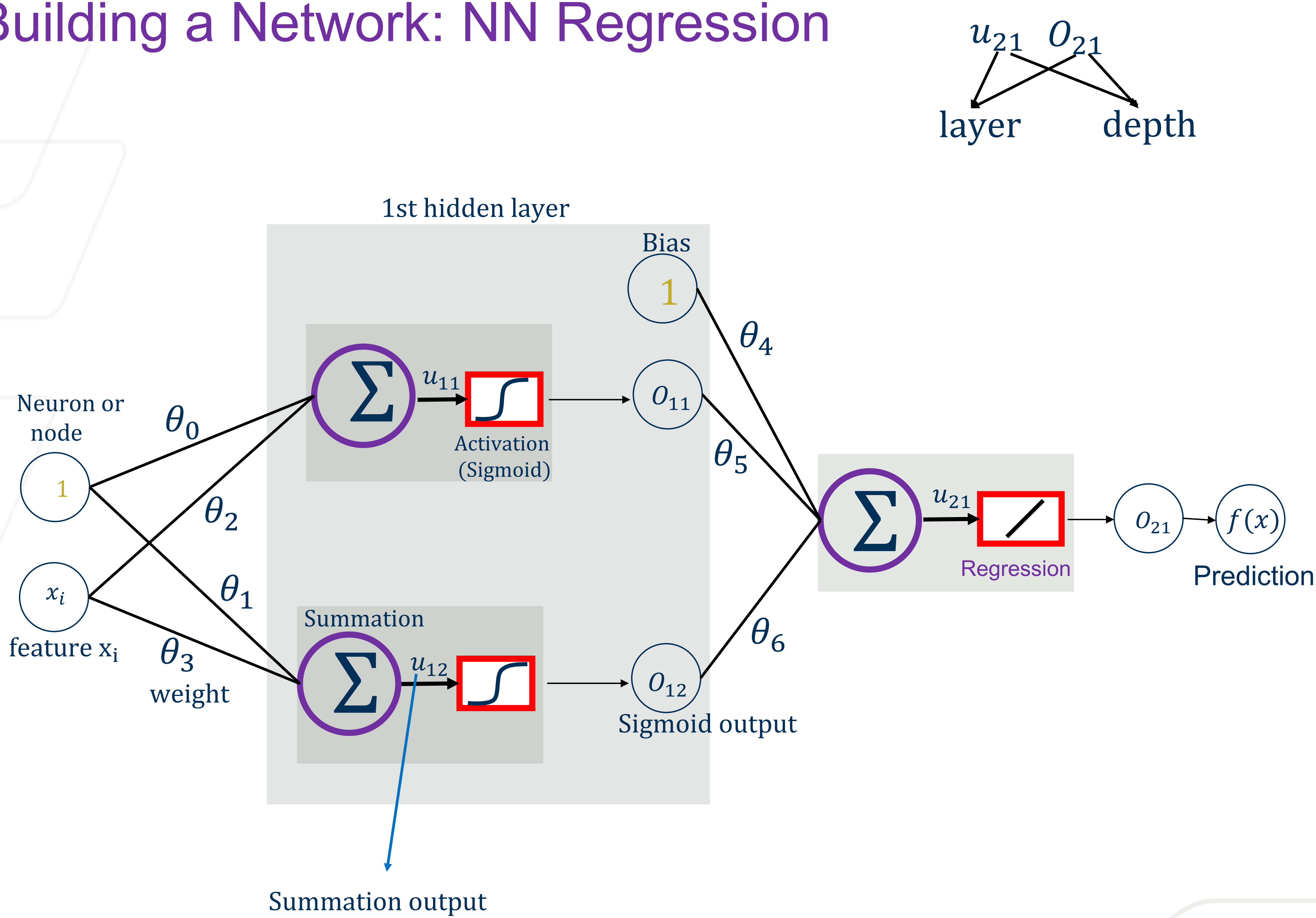
In Neural Network, 1 learning block is 1 building block

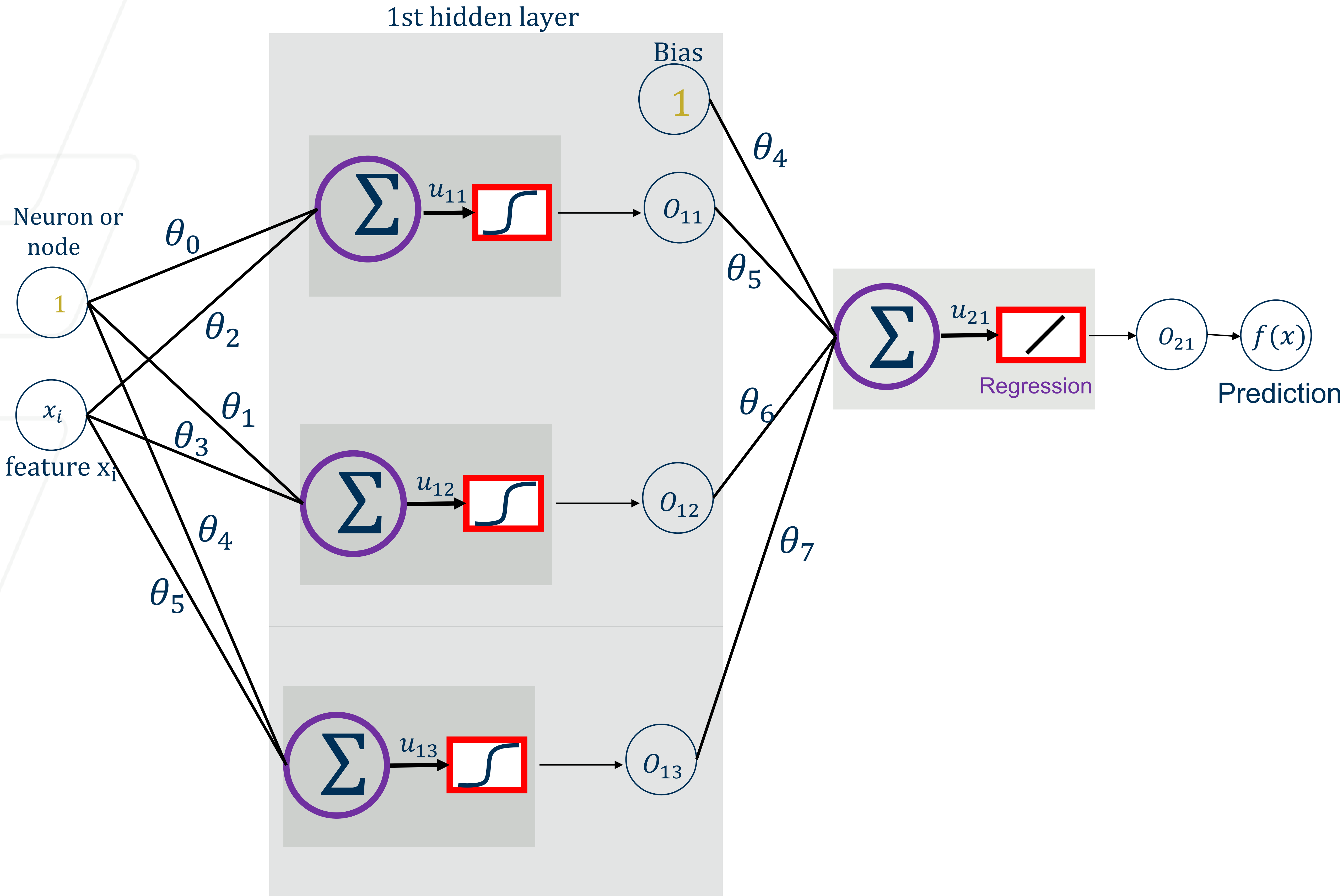


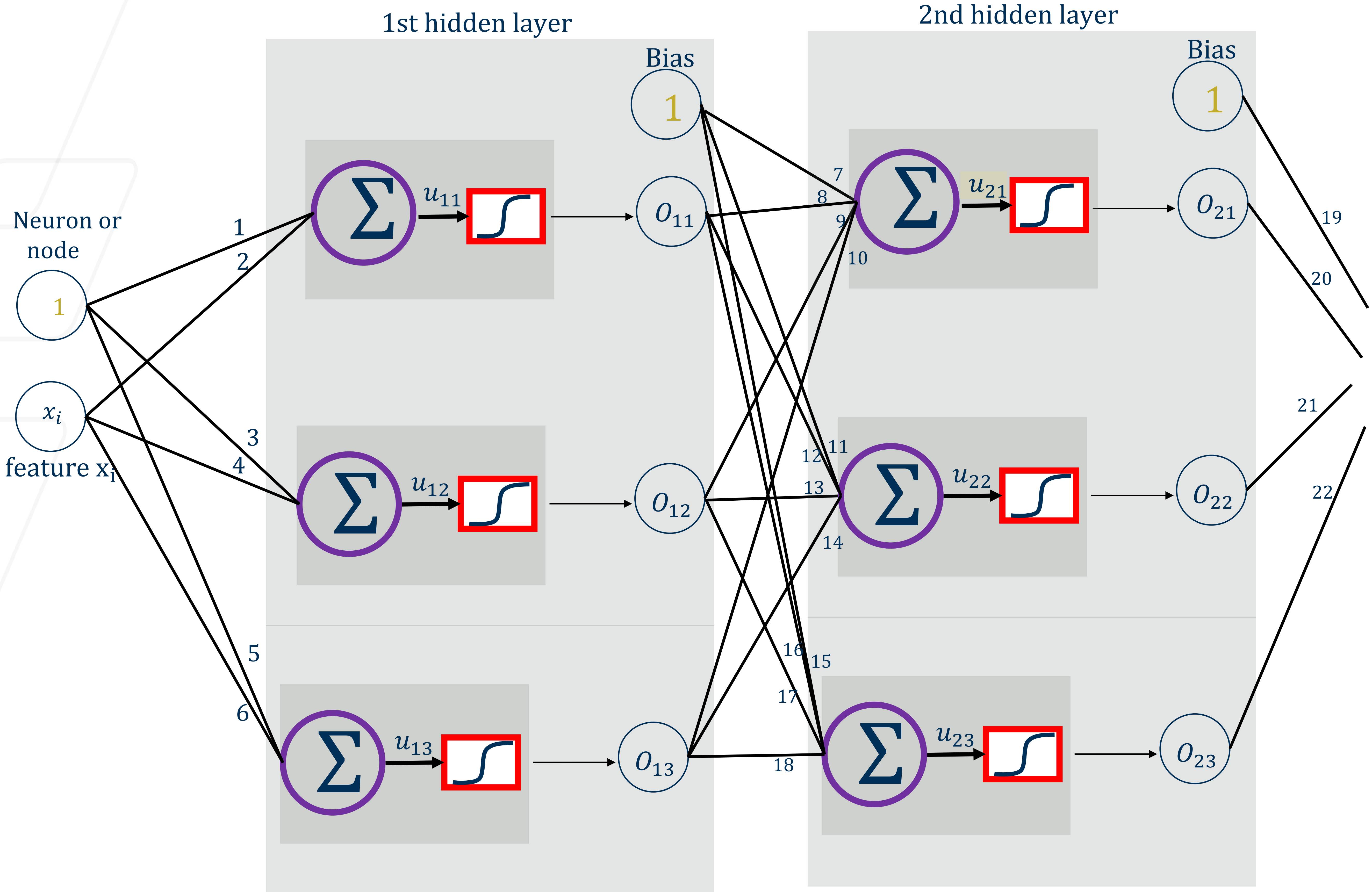
A **neural network** is essentially a **stack (or composition)** of many **learning blocks**, where:

$$\textit{Output of Block } i \;=\; \textit{Input to Block } (i+1)$$

Building a Network: NN Regression




1st hidden layer
Bias
1
θ_4
u_{11}
o_{11}
Neuron or node
θ_0
1
θ_5
θ_2
u_{21}
o_{21}
$f(x)$
Regression
Prediction
x_i
θ_1
θ_6
θ_3
feature x_i
u_{12}
o_{12}
θ_4
θ_7
θ_5
u_{13}
o_{13}


1st hidden layer
2nd hidden layer
Bias
Bias
Neuron or node
feature x_i
x_i
u_{11}
u_{12}
u_{13}
u_{21}
u_{22}
u_{23}
o_{11}
o_{12}
o_{13}
o_{21}
o_{22}
o_{23}
1
2
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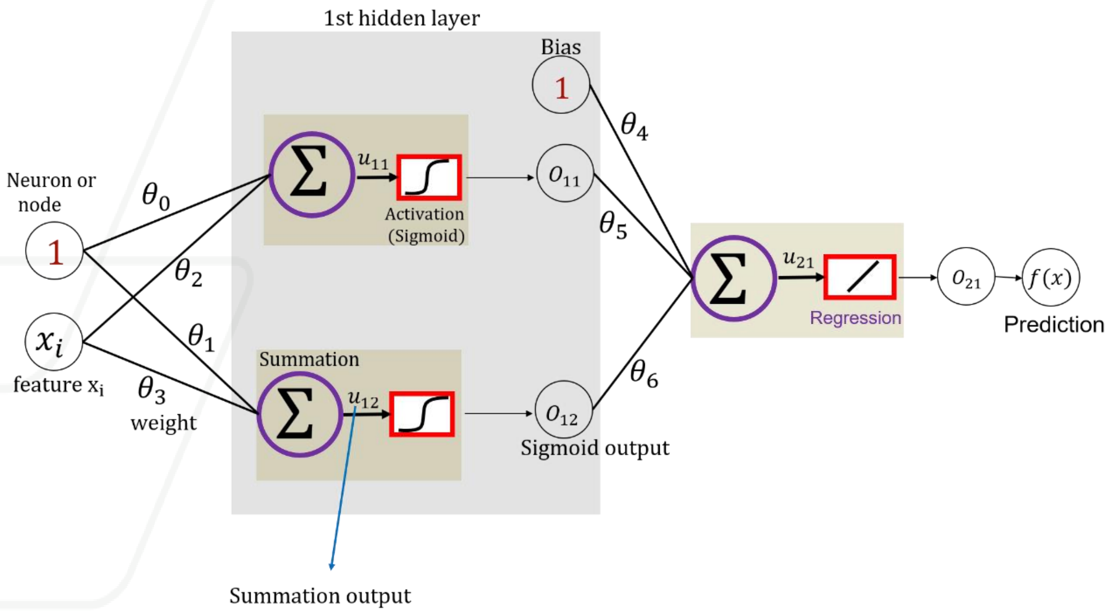
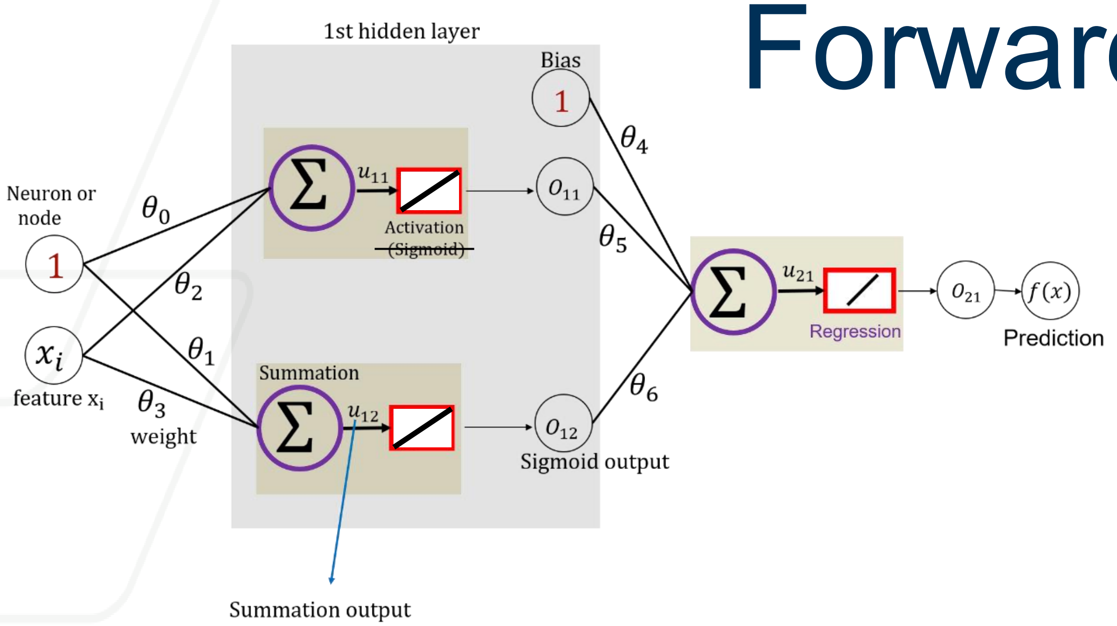
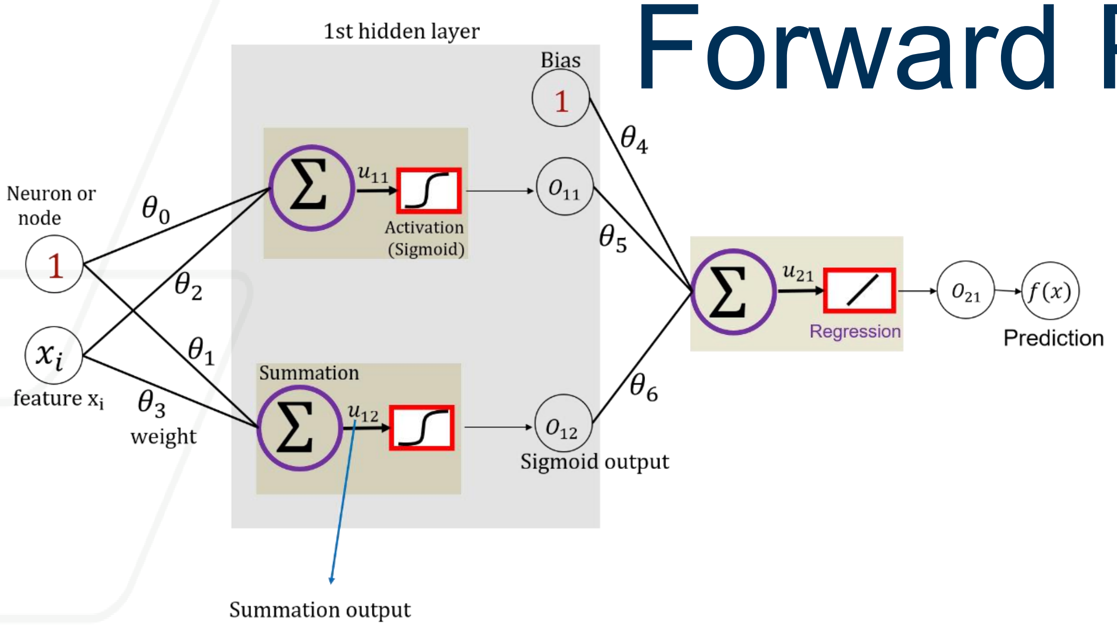
Gradient Descent



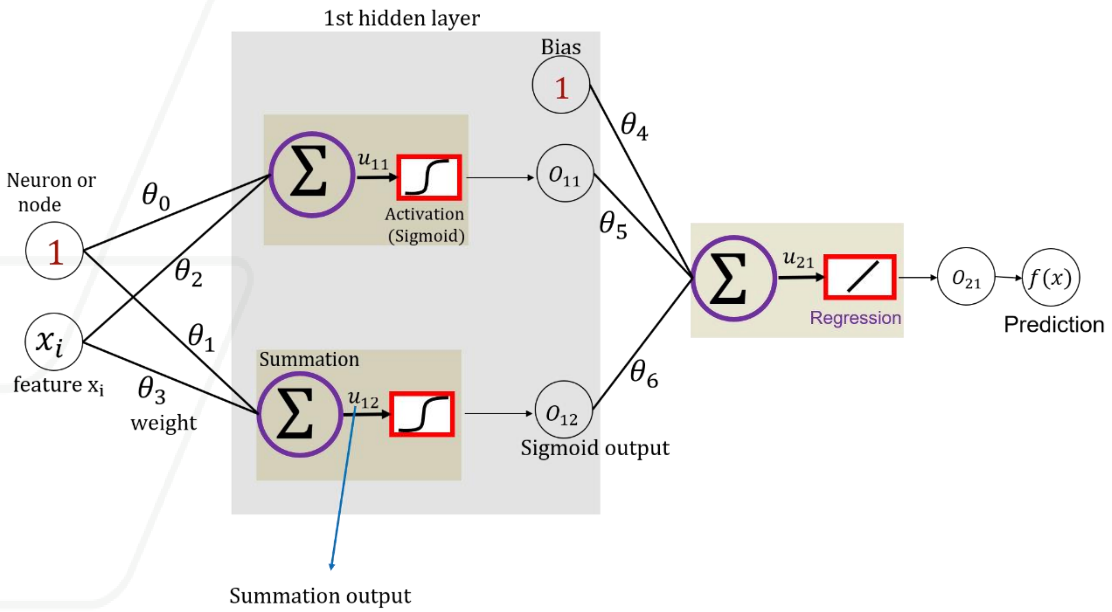
Forward Propagation: Rigid Network

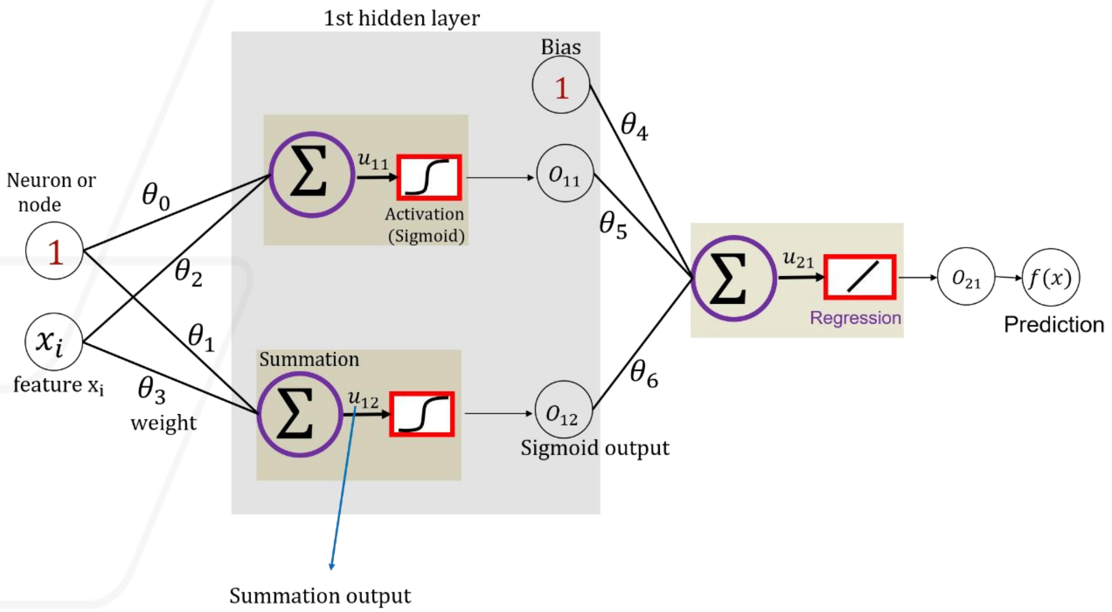


Forward Propagation: Flexible Network

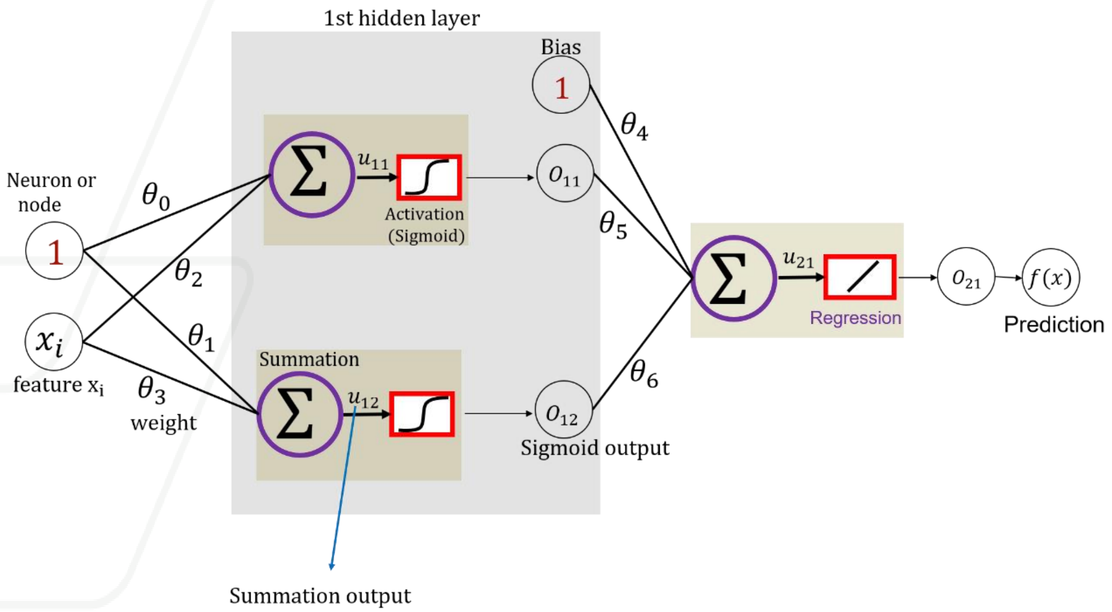


https://playground.tensorflow.org


1st hidden layer
Bias
1
θ_4
u_{11}
o_{11}
Activation
(Sigmoid)
Neuron or node
θ_0
1
θ_2
θ_5
u_{21}
o_{21}
$f(x)$
Regression
Prediction
x_i
θ_1
feature x_i
θ_3
weight
Summation
u_{12}
o_{12}
θ_6
Sigmoid output
Summation output



1st hidden layer
Bias
1
θ_4
Neuron or node
θ_0
u_{11}
Activation (Sigmoid)
o_{11}
θ_5
1
θ_2
u_{21}
o_{21}
$f(x)$
Regression
Prediction
x_i
θ_1
Summation
feature x_i
θ_3
weight
u_{12}
o_{12}
θ_6
Sigmoid output
Summation output



1st hidden layer
Bias
1
θ_4
Neuron or node
1
θ_0
u_{11}
Activation (Sigmoid)
o_{11}
θ_5
θ_2
x_i
θ_1
u_{21}
Regression
o_{21}
$f(x)$
Prediction
feature x_i
θ_3
weight
Summation
u_{12}
o_{12}
θ_6
Sigmoid output
Summation output